

Random Forest Evaluations

Full Model rf = RandomForestRegressor(n_estimators=300,
random_state=42,
max_depth=None,
min_samples_split=5,
min_samples_leaf=2,
n_jobs=-1,
verbose=1,
max_features=None
)

Computation time (~ 1 Hour)

Model: Random Forest Regressor

Features used: 113

Training samples: 176,592

Test samples: 44,148

Test Set Performance:

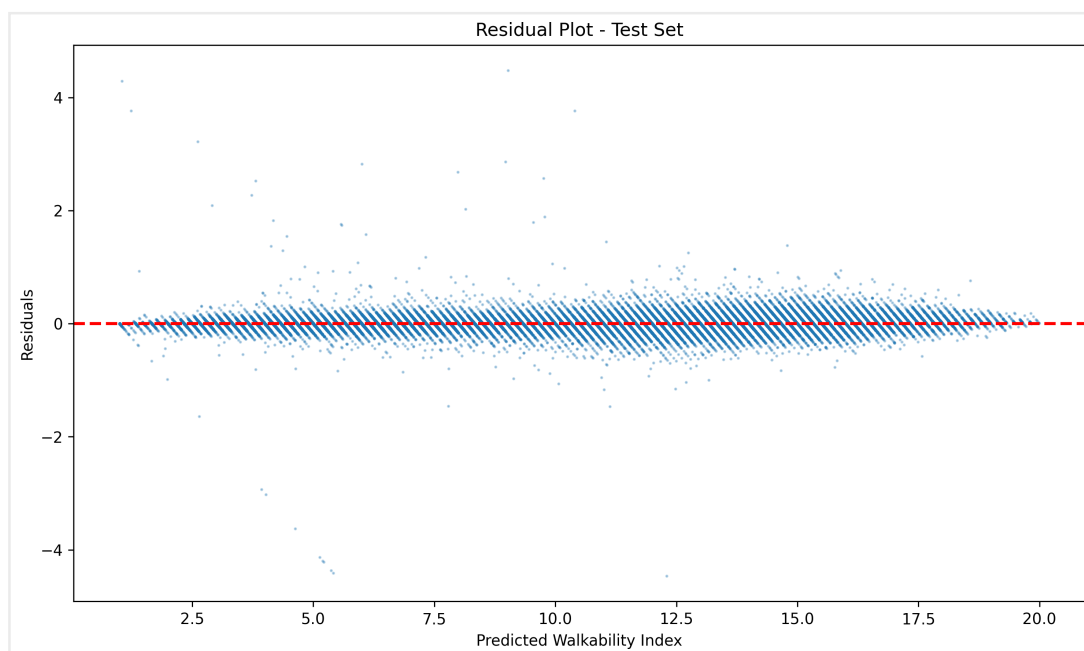
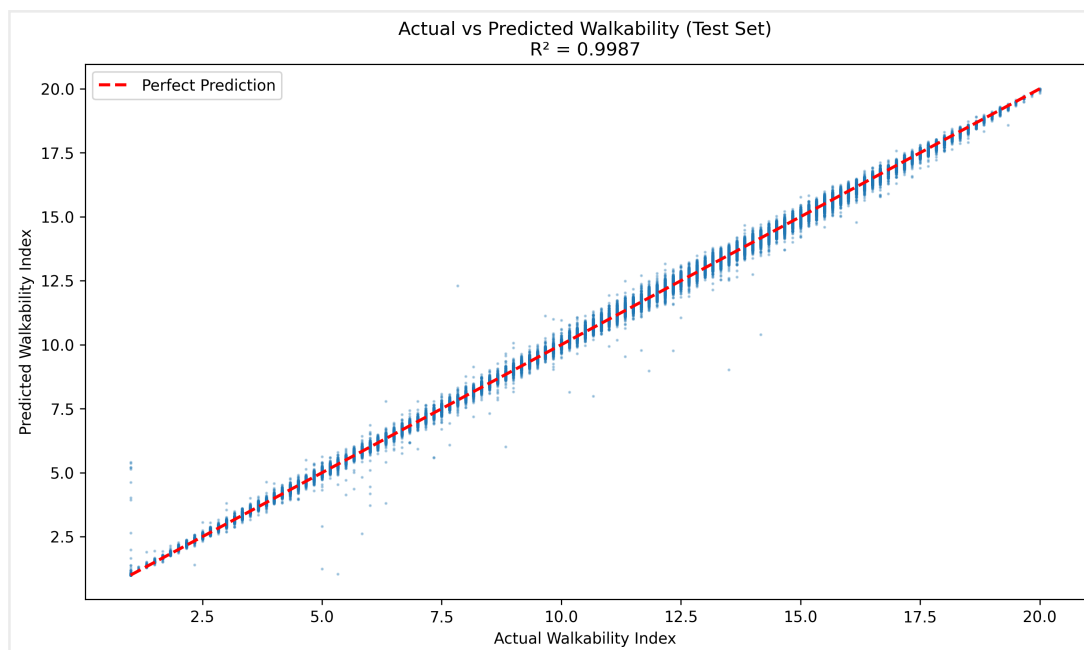
R² Score: 0.9987

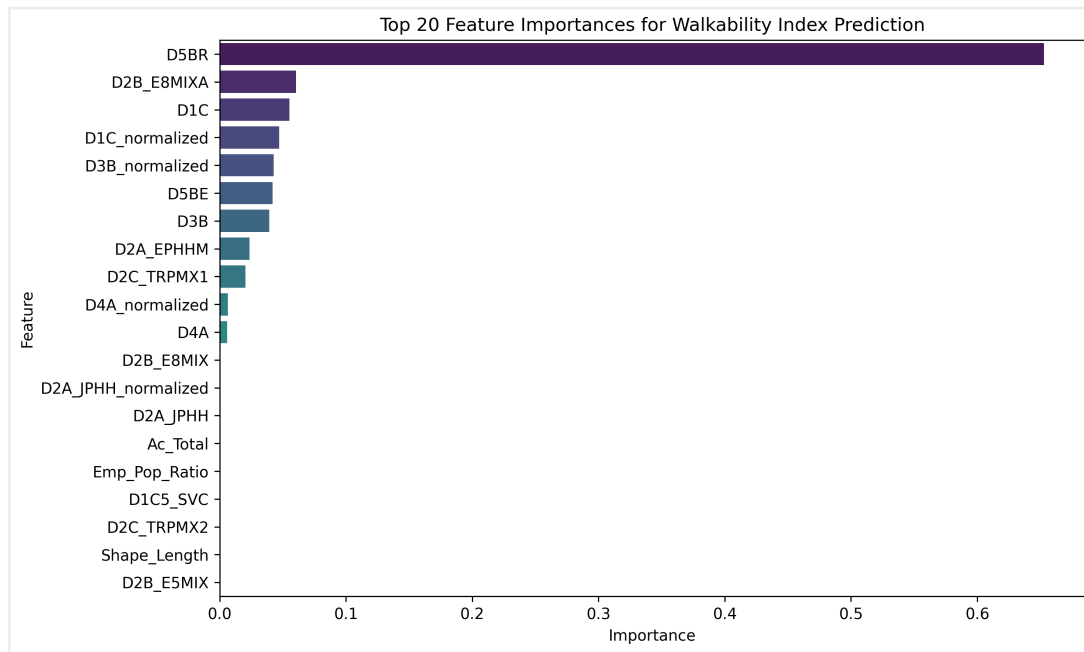
RMSE: 0.1574 (on 1-20 scale)

MAE: 0.0916 (on 1-20 scale)

Cross-validation R² scores: [0.99860324 0.9987155 0.99854225 0.99848413
0.99870748]

Mean CV R²: 0.9986 ± 0.0002





Full Model (Different parameters) `rf = RandomForestRegressor(n_estimators=100, random_state=42, max_depth=20, min_samples_split=10, min_samples_leaf=5, n_jobs=-1, verbose=1, max_features='sqrt')`

RESULTS

Computation time (1.3 mins)

Cross-validation R^2 scores: [0.99326341 0.9934971 0.99330943 0.99309633 0.99336401]

Mean CV R^2 : 0.9933 ± 0.0003

Model: Random Forest Regressor

Features used: 113

Training samples: 176,592

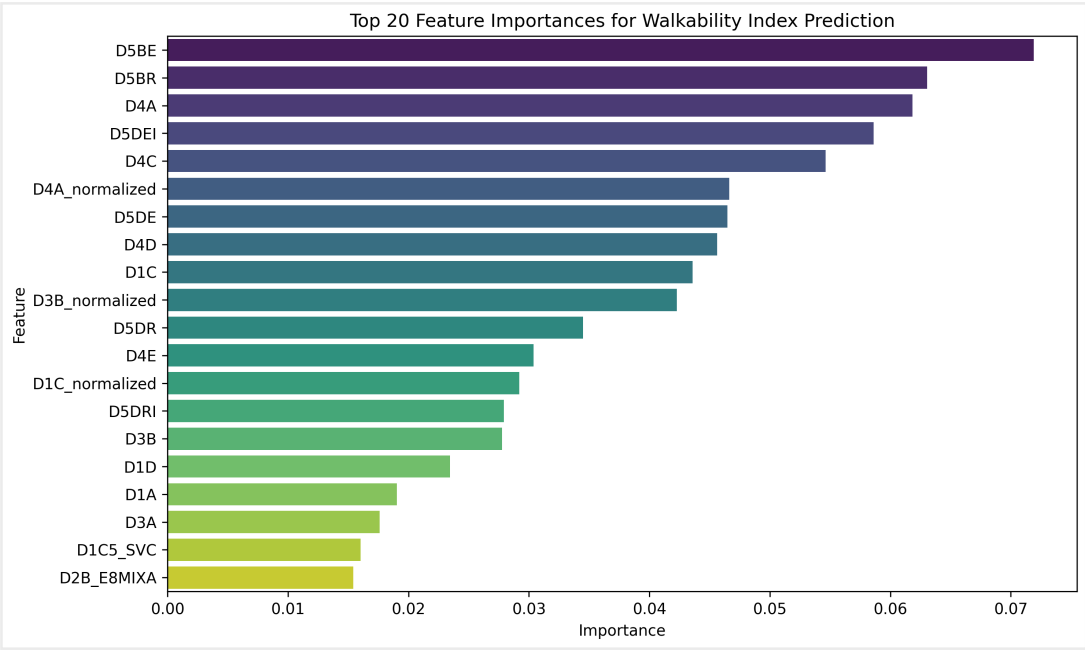
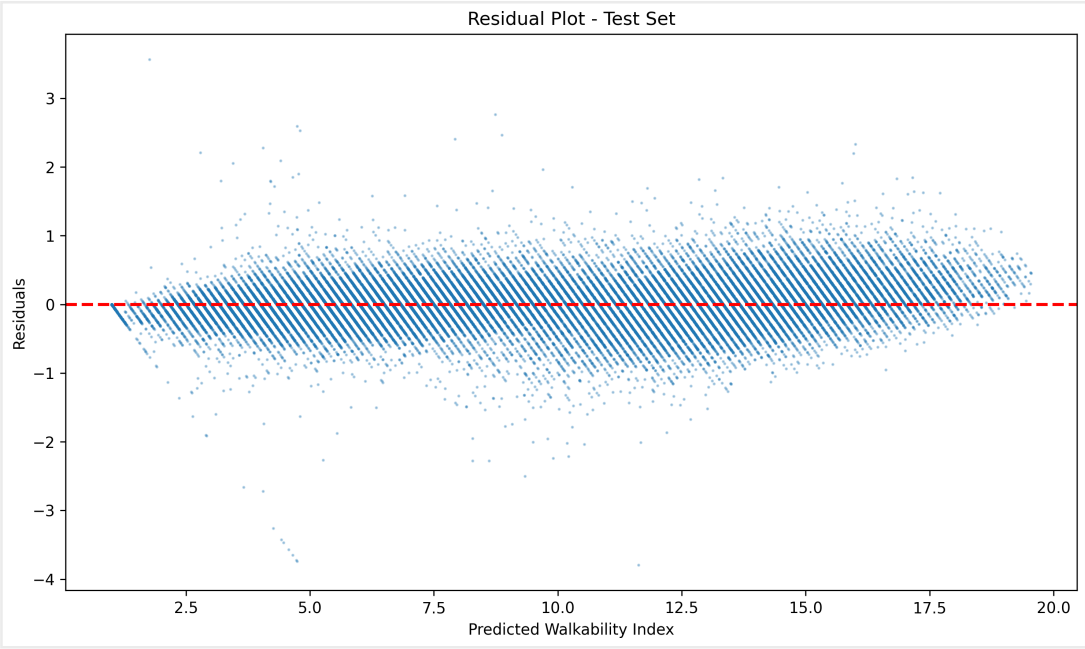
Test samples: 44,148

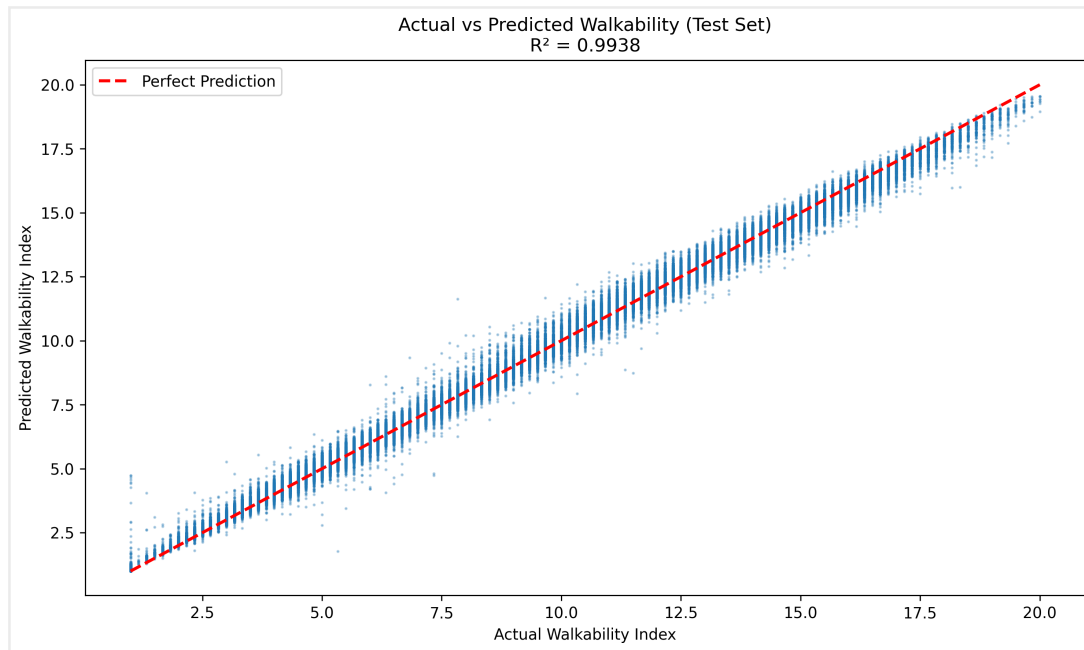
Test Set Performance:

R^2 Score: 0.9938

RMSE: 0.3451 (on 1-20 scale)

MAE: 0.2547 (on 1-20 scale)





USING OPTUNA FOR OPTIMIZING

Computation time (~30 mins) did 20 trials with 3 Fold CV

Optimized Model Summary:

Features: 20 (top features from previous analysis)

Optimization trials: 20

Best CV R^2 : 0.9929

Test R^2 : 0.9936

Test RMSE: 0.3494

Test MAE: 0.2446

Best hyperparameters:

n_estimators: 500

max_depth: 25

min_samples_split: 12

min_samples_leaf: 2

max_features: None

max_samples: 0.8663399949918865

